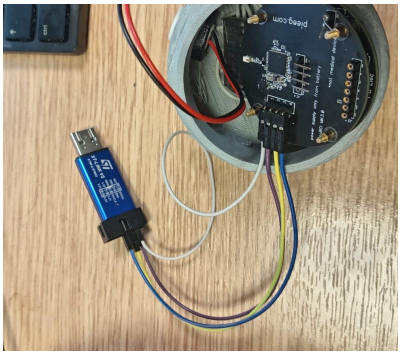
PCB & MANUFACTURING SPECIFICATION · XR · XR-16

BCI Module, Mask & Enclosure Materials

Bare-board specification of the two IronBCI-16 boards inside the PiEEG XR BCI box, the materials of the mask and the box, and the production quality test. Circuit design files are proprietary and are not part of this document.



FR-4	2 × 2	Ø 50	Ag/AgCl	Silicone
Board material	Boards × copper layers	mm round boards	Dry electrodes	Mask

01 BARE-BOARD FABRICATION PARAMETERS

Parameters for the two boards of the BCI module, as used by common PCB order forms. Each board is ordered separately.

Parameter	ADC board	MCU board
Base material	FR-4, standard Tg	FR-4, standard Tg
Layers	2	2
Outline	Ø 50.0 mm round	Ø 50.0 mm round
Thickness	1.6 mm *	1.6 mm *
Copper weight	1 oz (35 µm)	1 oz (35 µm)
Solder mask / legend	Black / white	Black / white
Surface finish	ENIG recommended (2 × TQFP-64)	**ENIG required** (QFN, 0201 parts)
Min. track / space	0.15 mm (6 mil)	**0.10 mm (4 mil)**
Min. drilled hole	Ø 0.40 mm	**Ø 0.30 mm**
RF note	—	Keep-out under the chip antenna
RoHS	Lead-free	Lead-free

* Thickness is not stated in the fabrication data; 1.6 mm is the nominal value used here.

02 MECHANICAL FEATURES OF THE BOARDS

Feature	Size	Qty	Board	Purpose
Mounting holes	Ø 3.00 mm	4 each	both	M2 spacers, 29.86 × 33.04 mm pattern
Electrode headers	Ø 1.00 mm plated	10 + 8	ADC	J3.3 (mask cable) and J3.4, 2.54 mm
Board-to-board headers	Ø 1.50 mm plated	7 + 2	both	SPI link and power
SWD / expansion	Ø 0.90 mm plated	4 / 5 / 4	MCU	Programming and expansion

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03 ASSEMBLY AND MATERIALS

BCI MODULE	
Technology	SMT on the top side of each board; through-hole headers
Key ICs	2 × TI ADS1299 (TQFP-64) on the ADC board · ST STM32WB55 (QFN-48) on the MCU board
Passives	0201 to 0805; 0201 on the MCU board needs a fine-pitch assembly line
RF	2.4 GHz chip antenna at the board edge; keep-out per the antenna datasheet
Firmware	BLE stack and the PiEEG XR application loaded over SWD (XR-SW-006)
FACE MASK	
Body	Silicone, moulded for one headset model, with a removable silicone insert
Electrodes	10 flat dry silver / silver-chloride (Ag/AgCl), pressed into Ø 1 mm holes
Cable	Multi-core cable in a braided sleeve, from the mask to the BCI box
BCI BOX	
Enclosure	Round 3D-printed plastic housing with a screw lid; no metal near the antenna
Power entry	USB-C socket, 5 V, wired to the module's 5 V input

04 PRODUCTION QUALITY TEST

1 Power
Power the box from a 5 V power bank through USB-C. An LED lights inside the box.

2 Bluetooth
The device advertises as `PiEEG XR`; PiEEG Cloud (PiEEG Devices → PiEEG XR) shows all channels at 250 SPS.

3 Shorted-input check
Before the mask is connected, short **REF** to each channel input. Every channel must read close to 0 µV.

4 Mask check
Connect the mask, wear it and blink hard 3–5 times: large spikes on the forehead channels Ch3–Ch6, every channel moving.

05 HANDLING, STORAGE AND COMPLIANCE

Topic	Requirement
ESD and storage	The module is ESD-sensitive: open the box only on an anti-static surface. Store dry, 10–35 °C.
Hygiene	Wipe the electrodes and silicone with an alcohol pad before and after each user; let them dry fully.
Classification	Research device with a 2.4 GHz BLE transmitter. Not a medical device; not certified for use on the human body.
Design files	Schematics, layouts, Gerbers and BOMs are proprietary to PiEEG and are not published.

06 RELATED DOCUMENTS

XR-DS-001 Product Datasheet	**XR-MD-002** Mechanical & Assembly Drawing	**XR-PIN-004** Mask, Electrode & Connector Guide
XR-QSG-005 Quick Start & Safety Guide	**XR-SW-006** Bluetooth Interface & Software Guide	

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